From Data Volume to Informational Noise: Configuration Complexity and Tool Siloing as the Main Causes of the Degraded Signal-to-Noise Ratio in SOCs

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Abstract

For several years, Security Operations Centers have faced a paradox that is increasingly hard to ignore. The more detection tools organizations deploy, the more the volume of alerts grows, and the harder it becomes for analysts to tell what really matters. This phenomenon, alert fatigue, is often blamed on an excess of data. Yet a careful reading of the literature and of the most recent surveys (2025-2026) suggests that volume is only a secondary factor. What truly degrades the signal-to-noise ratio is the way tools are configured and the degree to which they remain isolated from one another. This article proposes a framework to distinguish these effects and draws a few practical consequences.

Keywords: Security Operations Center, alert fatigue, signal-to-noise ratio, tool sprawl, configuration complexity, risk-based alerting.

1. Introduction

Fifteen years ago, a SOC often made do with a SIEM, a few sensors and a team of analysts who spent their days sorting logs. Today, most organizations run a stack of tools (EDR or XDR, threat intelligence platforms, GRC solutions, attack surface management modules, cloud tools, sometimes a SOAR), each generating its own stream of alerts. The figures are telling. The 2026 Microsoft/Omdia report indicates that 46% of alerts are false positives and that 42% are never looked at. The SANS SOC Survey of the same year shows that two thirds of teams say they can no longer keep up. Filigran, in its State of Threat Management, notes that nearly half of SOC experts believe their tools work poorly together.

Faced with these findings, one explanation comes up regularly: there is too much data. The idea is intuitive. Yet it does not survive scrutiny for long. In other fields that handle considerable volumes of information (quantitative finance, medical imaging, logistics), increasing volume, when accompanied by careful integration and relevant configuration, generally improves the quality of decisions. Why would cybersecurity be an exception?

The thesis I defend here is more precise. The noise that is suffocating contemporary SOCs does not primarily arise from the volume of data. It arises from the configuration complexity of the tools and from their siloing. In other words, it is not so much the data that creates the noise as the way it is produced, configured and isolated.

2. What the Literature Says

Alert fatigue is not a new concept. As early as the mid-2010s, several studies had already described the gradual desensitization of analysts facing an almost constant flow of low-relevance alerts. More recent surveys only confirm the scale of the problem. A combined SANS, ISC2 and partners study (2025) already indicated that 66% of SOC teams could no longer absorb the volume and that 90% were accumulating significant backlogs. The same body of work showed that nearly 44% of daily alerts were never investigated and that more than 70% of analysts reported symptoms of burnout. These figures have barely moved since, despite massive investment in automation.

The signal-to-noise ratio remains the most useful indicator for describing the situation. The signal consists of events genuinely linked to a threat or an incident. The noise includes false positives, duplicates, alerts without context and, more broadly, anything that consumes analyst time without providing actionable information. Several literature reviews, notably that of Ndichu et al. published in 2026, show that this ratio has continuously deteriorated between 2018 and 2026.

Tool sprawl explains part of this deterioration. The average enterprise now manages between eleven and eighteen

detection and response tools, sometimes more. Each solution brings its own detection logic, thresholds, taxonomy and often its own interface. A single suspicious behavior (a brute-force attempt, lateral movement, an identity anomaly) can then generate several distinct alerts that do not talk to each other. The analyst ends up manually rebuilding a chain of events that could have been correlated upstream. The 2026 Microsoft/Omdia and Vectra studies confirm that the most fragmented organizations are also those that leave the most alerts unaddressed.

The configuration of detection rules plays at least as important a role. Most tools ship with "out-of-the-box" configurations designed to minimize false negatives. The price to pay is a high rate of chronic false positives. Fraunhofer FKIE's work on risk-based alerting shows quite clearly that the severity declared by the tool alone is a poor predictor (mean AUROC around 0.72). By contrast, risk scores that combine several dimensions (alert volume on a single asset, diversity of alert types, business criticality) achieve markedly better performance (mean AUROC of 0.92). This suggests that the way value is assigned to alerts matters more than their sheer number.

The solutions proposed in recent years (risk-based alerting, XDR platforms, SOAR, agentic AI-based approaches, continuous threat exposure management) mostly address the downstream handling of alerts. They help sort faster or correlate after the fact. They rarely address upstream complexity, that is, the way tools are configured and connected to each other. This is precisely the point that remains under-explored in the literature.

3. A Simple Framework for Clarity

Three factors that contribute to noise in a SOC can be distinguished:

- the volume of data generated;
- configuration complexity (number of rules, heterogeneity of thresholds and taxonomies, lack of adaptation to the business context);
- the degree of siloing (lack of integration and native correlation between tools).

Volume acts mainly as an amplifier. A high volume combined with careful configuration and good integration can produce a perfectly acceptable signal-to-noise ratio. Conversely, even a moderate volume quickly becomes noisy once tools are poorly configured and isolated from one another. This can be summarized as follows: effective noise depends on the product of volume and the sum of configuration complexity and siloing.

This model has the advantage of simplicity. It helps explain why some organizations, despite facing large volumes, fare better than others: they have invested in detection engineering and integration rather than in the permanent addition of new alert sources.

4. What Recent Data Show

The 2025 and 2026 surveys converge on several points. Alert volume has indeed increased, sometimes by more than 50% in a year in environments that have massively adopted AI tools. But the organizations that complain most about alert fatigue are also those that report the largest number of non-integrated tools and the lowest level of contextual configuration. Work on risk-based alerting shows that better prioritization significantly reduces the false-positive burden, even at constant volume. Finally, companies that have rationalized their stack or seriously invested in detection engineering report clearer improvements in the signal-to-noise ratio than those that merely added downstream processing capacity.

In other words, volume amplifies the effects of poor configuration and heavy siloing. It is not the primary cause.

5. Discussion

The idea that "too much data creates noise" has the merit of simplicity. It also has the drawback of diverting attention from the levers that are actually actionable. What produces noise is, first of all, three concrete mechanisms.

The first is duplication. A single technical event triggers independent alerts in several tools. The analyst must then manually reconstruct a chain that could have been correlated upstream. The second is loss of context. A technically correct alert loses much of its value if it is not enriched with the context of the asset, the user's history or threat intelligence information. Siloing precisely prevents this enrichment. The third mechanism concerns thresholds and

rules. Default configurations favor detection at the cost of a high false-positive rate. Without continuous detection engineering work, this trade-off deteriorates over time.

Purely technical solutions centered on downstream triage (more AI, more playbooks) may relieve teams in the short term. They do not address the cause. A lasting reduction in noise requires acting on the two multiplicative factors of the model: configuration complexity and the degree of siloing. Risk-based alerting and continuous threat exposure management approaches go in this direction, provided they are properly implemented. Vigilance is also needed: some poorly calibrated generative AI solutions can themselves become a new source of noise by producing alerts that are poorly contextualized or hard to explain.

6. Some Implications

For operational teams and CISOs, several avenues emerge. It is useful to regularly measure the signal-to-noise ratio per tool and per use case, rather than settling for the overall alert volume. Before adding a new tool, it is often better to rationalize those that already exist. Investment in contextual detection engineering generally pays off more than adding triage capacity. Risk-based alerting and continuous threat exposure management approaches deserve to be explored, provided that quality input data is available.

On the research side, work remains. Standardized metrics of “effective noise” that go beyond the simple false-positive rate are still lacking. Longitudinal studies measuring the real impact of tool rationalization and improved configuration would be particularly useful. Finally, agentic solutions must be evaluated under realistic operational conditions, and not only on public datasets.

7. Conclusion

The noise affecting contemporary SOCs is not an inevitability tied to data volume. It results above all from configuration complexity and tool siloing. By refocusing efforts on integration, detection engineering and risk-based prioritization, it is possible to improve the signal-to-noise ratio without giving up detection coverage. This clarification invites us to move beyond the somewhat too simple “too much data” hypothesis and to look at the concrete mechanisms that turn telemetry into noise.

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